


Technical Design Document (TDD)

# F1tenth PoliMI

## The car: stack, car interface and deployment

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## Document History

| Version | Date | Author | Changes |
|---|---|---|---|
| 0.01 | 2026-10-25 | F1tenth PoliMI team | Initial draft from the car-interface design proposal and the runbooks; `car.md` and the hardware studies folded in (section A) |
| 0.02 | 2026-11-08 | F1tenth PoliMI team | Review of v0.01: DEC-08 follows the Jazzy migration; DEC-12 allows our packages in a fork's own folder; JetPack 7.2.1 on the car (DEC-13); Jazzy migration summarised (section 7.3); stop-test chain, steering limits (T-12) and the `/joy` stand-in owner; references to working notes removed |
| 0.03 | 2026-11-09 | F1tenth PoliMI team | Controllers and tuning topics (PID and Pure Pursuit, FR-PTH-009); camera calibration procedure in the car bringup (FR-DRV-005, T-06); line detection on replayed camera bags; IMU model deferred until OI-08 |
| 0.04 | 2026-11-09 | F1tenth PoliMI team | Jazzy migration step 1 done: the images run on `ros:jazzy` with the DEC-02 packages and `f1tenth_system` on `jazzy-polimi`; today-versus-target, deployment and repositories updated |

## Document Approvals

| Role | Name | Date | Signature |
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| Technical Lead | | | |
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| Management | | | |

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# 1 Introduction

## 1.1 Purpose

This document defines how the car's software is built: the layers of the stack, the car interface the stack talks to, how the stack is started, where the car's measured numbers live, and how it is deployed on the car and on a developer's machine.

## 1.2 Scope

- **In scope:** the stack on the car, the car interface (topics and frames), the car's hardware layer, the repositories, the Docker images, the development standards.
- **Out of scope:** the simulator's internals (the gym, the bridge, `polimi_sim`). Here the simulator appears only as the second implementation of the car interface (`hw:=sim`). Its design is in `TDD_Gym`.

## 1.3 References

- `RASD_F1tenth` — requirements refined here.
- The README (first-time setup), `CONTRIBUTING.md` (git workflow, access, rulesets) and the runbooks `docs/software/docker.md` and `simulator.md`.
- `TDD_Gym`, the simulator's design.
- Hardware studies: LiDAR (23 Sep 2026), camera (7 Oct 2026), chassis (24 Sep 2026), folded into section A.

## 1.4 Status Conventions

**[Decided]** agreed **[Proposed]** proposed, waiting for the team's confirmation **[V2]** later phase **[TBD]** open.

> **Today versus target.** The images run ROS 2 ~~Humble today~~ Jazzy and the drivers come from ~~`humble-polimi`. Jazzy is decided (DEC-01) and the migration is planned~~ `jazzy-polimi`: step 1 of the Jazzy migration (section 7.3) ~~, not done.~~ is done. The simulator still runs the Humble-era gym and bridge, and `polimi_sim` is still in this repo, until steps 2 and 3. This document describes the Jazzy target; where today differs, it says so.

# 2 Architecture

> **Key point.** One stack, two cars. The stack talks to the car only through a fixed set of ROS topics and TF frames, the **car interface**. On the car, `f1tenth_system`'s drivers and `orbbec_camera` serve it. In the simulator, our fork of `f1tenth_gym_ros` serves the same interface. The stack cannot tell which one it is running on, and neither fork depends on anything of ours. **[Proposed]**

## 2.1 Component View

Three layers. The logic and the common layer are one stack and run unchanged everywhere. The hardware layer is exactly one of two, picked by the launch argument `hw:=car` or `hw:=sim`.

The seam sits below the mux, at `/ackermann_cmd`, the topic the VESC reads. The RASD requires joystick takeover, the deadman and AEB to work on the car and in the simulator (FR-SIM-002), so the mux and the safety nodes belong to the common layer, not to the hardware.

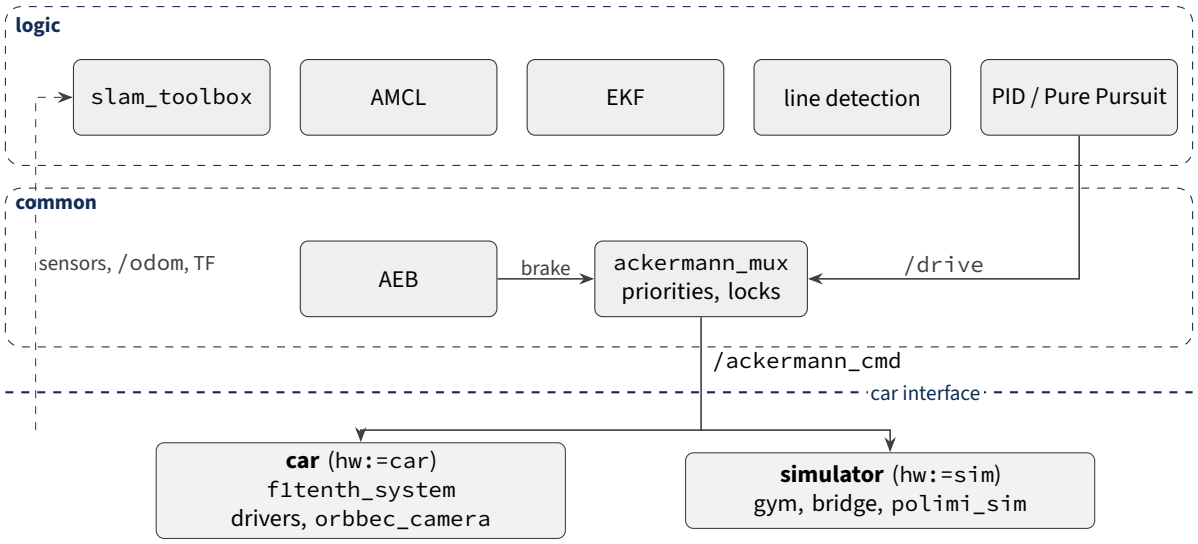



**Figure 1:** Layers. Only `/ackermann_cmd` goes down through the car interface; sensors, odometry and TF come up with the same names whichever hardware runs.

## 2.2 Components

| Component | Responsibility | Interfaces |
|---|---|---|
| *Logic* | | |
| `slam_toolbox` | Map the room (FR-MAP) | `/scan`, TF → `/map` |
| AMCL | Pose in the room map (FR-LOC) | `/scan`, `/odom`, `/map` → TF `map`→`odom` |
| EKF (`robot_localization`) | Fused `odom` **[TBD: OI-08]** | `/odom`, IMU → TF `odom`→`base_link` |
| `polimi_lines` | Line detection (FR-LIN) | camera image, info → `nav_msgs/Path` |
| `polimi_control` | PID and Pure Pursuit, same interface (FR-PTH) | path, TF → `/drive`, tracking errors |
| *Common* | | |
| `ackermann_mux` | Pick the highest-priority input still sending; locks (FR-SAF) | `/teleop`, `/drive`, `/joy` → `/ackermann_cmd` |
| `polimi_safety` | AEB (FR-SAF-004) | `/scan`, `/odom` → mux input |
| `polimi_bringup` | The car interface (README), launches, our configs | `stack.launch.py hw:=car\|sim` |
| *Hardware, car* | | |
| `vesc_driver`, `vesc_ackermann` | Motor, servo, odometry (FR-DRV-001) | `/ackermann_cmd` → VESC; VESC → `/odom` |
| `joy`, `joy_teleop` | Joystick (FR-DRV-002) | `/joy`, `/teleop` |
| `orbbec_camera` | SL450 LiDAR, Gemini 335L camera and IMU (FR-DRV-003) | `/scan`, image, info, IMU |
| *Hardware, simulator* (see `TDD_Gym`) | | |

| Component | Responsibility | Interfaces |
|---|---|---|
| fork of `f1tenth_gym_ros` | Serves the same interface | as the car |

## 2.3 Runtime View

### 2.3.1 Command path

Autonomy publishes `/drive`. A person drives through `/teleop`, from the joystick on the car or the keyboard in the simulator. The mux picks by priority (joystick 100, autonomy 10, as in today's `mux.yaml`); the safety locks of OI-09 sit in the same mux. Below the seam, the car turns `/ackermann_cmd` into VESC commands; the simulator turns it into gym actions through its actuation model.

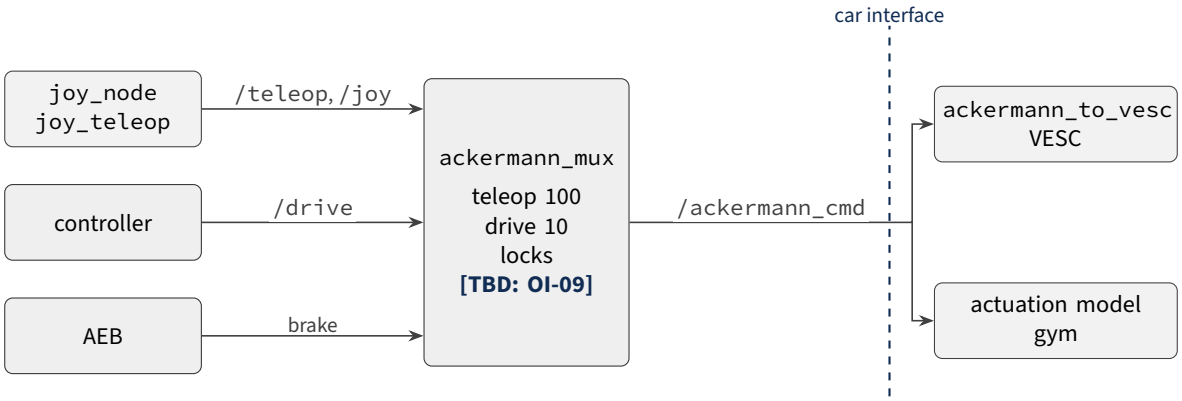



**Figure 2:** Command path. Grey boxes are hardware-specific: in the simulator a keyboard bridge publishes `/teleop` in place of the joystick nodes, and the car's VESC is replaced by the actuation model. Everything else is identical on car and simulator, so takeover, deadman and AEB are tested in the simulator as they run on the car.

### 2.3.2 Sensor path

Each topic in the middle column is part of the car interface. The car and the simulator each have one producer per topic; the stack subscribes only to the middle column. The simulator's ground truth stays out of it: it is for debugging and scoring only, never read by the stack (FR-SIM-003).

## 2.4 Deployment View

### 2.4.1 Machines and images

One Docker image per machine type, built from `docker/ros/Dockerfile`. The repo's code, configs and maps are bind-mounted, so code is edited on the host and run in the container; `build/`, `install/` and `log/` live in Docker volumes.

| Service | Machine | Simulator | Networking |
|---|---|---|---|
| `arm` | Apple Silicon Mac | yes | port 8765 (Foxglove), bound to 127.0.0.1 |
| `x86` | Linux or Windows PC | yes | port 8765 (Foxglove), bound to 127.0.0.1 |

| | | | |
|---|---|---|---|
| `jet` | Jetson Orin Nano on the car | no | host |
| `novnc` | optional, RViz in the browser | – | port 8080 |

Base image `ros:jazzy` on Ubuntu 24.04 (DEC-01)~~; today `ros:humble`~~, chosen by the build argument `ROS_DISTRO`. On the car the host is JetPack 7.2.1, also Ubuntu 24.04, so the container can use the GPU (DEC-13). Middleware `rmw_zenoh_cpp`; the Zenoh router starts with the container.

### 2.4.2 The car

| **Device** | **Link to the Orin Nano** | **Driver** |
|---|---|---|
| Orbbec Pulsar SL450 | Ethernet (the only port) | `orbbec_camera`, `lidar.launch.py` |
| Orbbec Gemini 335L | USB 3 | `orbbec_camera`, `gemini_330_series.launch.py` |
| VESC 6 MkVI | USB, `/dev/sensors/vesc` (udev rule **[TBD]**) | `vesc_driver` |
| 8BitDo Ultimate 2C | wireless dongle, USB | `joy` |

### 2.4.3 Repositories

| **Repo (`tfc-org`)** | **Submodule path** | **Holds** |
|---|---|---|
| `f1tenth_polimi` | – (this repo) | Our packages, configs, maps, Docker, docs |
| `f1tenth_system_polimi` | `ros2_ws/src/f1tenth_system` | Upstream drivers, branch `jazzy-polimi` ~~(today `humble-polimi`)~~ |
| `f1tenth_gym_polimi` | `ros2_ws/src/f1tenth_gym_ros` (after the migration) | Upstream bridge plus `polimi_sim`, branch `dev-jazzy-polimi` |

Dependencies point one way only: from our repo outwards. Neither fork knows our stack exists (DEC-08).

### 2.4.4 Launch

One entry point, `stack.launch.py`, with `hw:=car` or `hw:=sim`. It always starts the common layer and the logic asked for, plus exactly one hardware launch. **[Proposed]**

- `stack.launch.py hw:=car|sim`
  - `common.launch.py`: `ackermann_mux` with our `mux.yaml`, safety.
  - `car.launch.py` (`hw:=car`): `f1tenth_system`'s driver nodes one by one, `orbbec_camera`, the mount TF from `config/car/`.
  - `sim.launch.py` (`hw:=sim`): includes the fork's `sim_car.launch.py` with `config_dir:=~/ws/config` and, from DEC-08, `teleop_topic:=teleop`; nothing else.
  - the logic: SLAM or localization, line detection, controller, as asked.

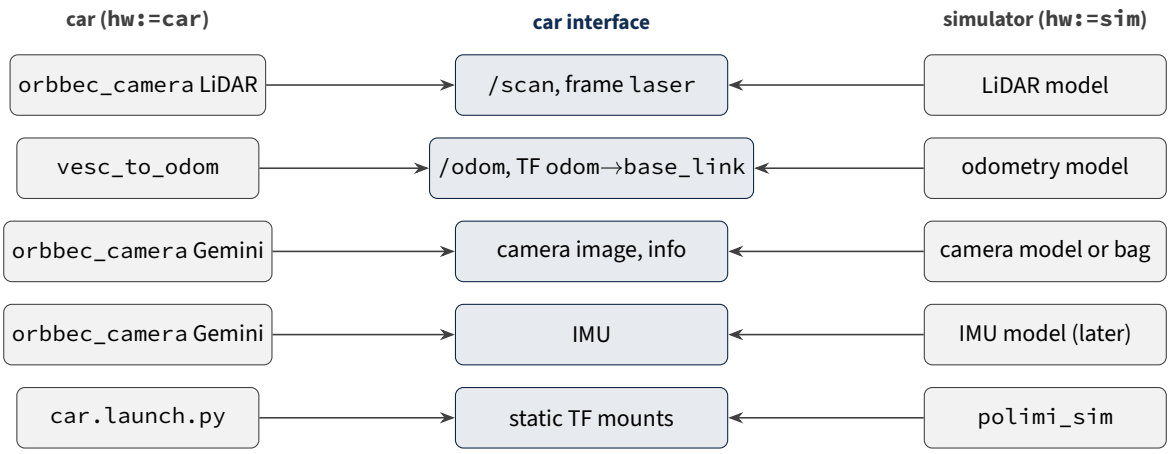



**Figure 3:** Sensor path. Same topic, same type, same frame, two producers. The mount TF comes from the same measured files on both sides (DEC-09), so the simulator places the LiDAR where the car's TF says it is.

Today the car starts with `f1tenth_stack`'s `no_lidar_bringup_launch.py` and the simulator with `simcar`, which becomes a shortcut for `hw:=sim`.

## 2.5 Design Decisions

| ID | Decision | Rationale / alternatives |
|---|---|---|
| DEC-01 | ROS 2 Jazzy on every image **[Decided]** | Humble ends on 31 May 2027, before or during the summer school; Jazzy is supported to 31 May 2029. Lyrical is too new: no `f1tenth_gym_ros` branch, no JetPack for its Ubuntu. (OI-13, 21 Oct 2026.) |
| DEC-02 | One image rebuild adds `nav2_amcl`, `robot_localization`, `image_transport_plugins`, `cv_bridge` **[Decided]** | In the Jazzy base apt layer, so AMCL and the EKF can be tried before any stack package exists. `cv_bridge` brings apt's `python3-opencv`; the apt layer goes first and pip after, so the gym's pip `cv2` and NumPy shadow apt's. Python nodes never import `cv_bridge` (its Python module breaks under the gym's NumPy 2) and convert images with NumPy; C++ nodes are unaffected. A `numpy<2` pin is no fallback, as the JAX gym needs NumPy $\geq$ 2.2: if a Python node ever needs `cv_bridge`, build it from source against NumPy 2. Checked after the rebuild (section 7.3, step 1). (OI-10, 22 Oct 2026.) |
| DEC-03 | Seam below the mux, at `/ackermann_cmd` **[Proposed]** | The simulator replaces the VESC, the motor, the chassis and the sensors, nothing above them. With the mux above the seam, takeover, deadman and AEB are the same nodes in both places. (R1) |
| DEC-04 | Plain topics are the contract; no extension mechanism **[Proposed]** | Our launch includes the fork's launch with two arguments; one switch, `hw:=`. No new code in the fork, and the contract can be checked with `ros2 topic info` on both sides. (R2) |

| ID | Decision | Rationale / alternatives |
|---|---|---|
| DEC-05 | `polimi_bringup` holds the contract and the launches; one package per function for the logic **[Proposed]** | Functions have different dependencies (the camera stack versus none) and their own tests; one bringup package keeps how it all starts in one place. (R3) |
| DEC-06 | Our own `car.launch.py`, not `f1tenth_stack`'s bringup **[Proposed]** | Upstream's `bringup_launch.py` starts the Hokuyo driver, its own mux and a hard-coded laser TF, none of which match our car. `f1tenth_system` stays a pure upstream fork: the scope's "our fork, extended with all our drivers" (`RASD_F1tenth` C-022) becomes "its driver packages, launched by our bringup". (R4) |
| DEC-07 | Our own `mux.yaml`, in `polimi_bringup/config/` **[Proposed]** | Starts from today's values (joystick 100, autonomy 10). OI-09 adds the e-stop and deadman locks and the AEB input, and those are ours, not upstream's. (R5) |
| DEC-08 | The simulator fork runs from a bare clone, after the Jazzy migration **[Proposed]** | During the migration the fork keeps `ackermann_mux` in `sim_car.launch.py` and needs `f1tenth_system` next to it, so the migration changes only the distro, the gym and the repo layout (8 Nov 2026). Afterwards, with `polimi_bringup`'s `common.launch.py`: `polimi_sim` drops the mux node and its `ackermann_mux` and `f1tenth_stack` dependencies, and gains `teleop_topic` (default `ackermann_cmd`). Earlier, the simulator would have no mux until the common layer exists. (R6) |
| DEC-09 | The hardware side publishes the mount TF, from the same files **[Proposed]** | Car: `car.launch.py` from `config/car/*.yaml`. Simulator: `polimi_sim` from the same files via `config_dir`. One file, two readers. (R7) |
| DEC-10 | `odom`→`base_link` has one owner at a time **[Proposed]** | The odometry publishes it until the EKF lands (OI-08); then the EKF does, and `publish_tf` goes off on car and simulator in the same change. Two publishers make the tree jump. (R8) |
| DEC-11 | Everything runs in Docker, code bind-mounted **[Decided]** | Same images on Mac, PC and car; the host needs only Docker and git. |
| DEC-12 | In a fork, upstream's files take only upstreamable fixes; our code lives in `polimi_*` packages, in our repo or in a fork's own folder **[Decided]** | Upstream fixes can be pulled and offered back, one commit each; only fixes we cannot make from outside touch upstream's files. `polimi_sim` lives in the simulator fork's `polimi_sim/` folder, never mixed with upstream's files (RASD NFR-MNT-001). |
| DEC-13 | JetPack 7.2.1 on the Orin Nano **[Decided]** | Jetson Linux 39.2.1, Ubuntu 24.04 like the `ros:jazzy` containers, CUDA 13: the GPU is usable from the container (RASD C-024), for example by a learned line detector (OI-06). JetPack 6 is Ubuntu 22.04, where a 24.04 container runs on the CPU only. (8 Nov 2026.) |

# 3 Data

The measured car lives in `ros2_ws/config/`, written once and read by both the car and the simulator. The simulator fork's defaults only fill gaps and never hold our measurements; it logs where each value came from.

| File | Readers | Holds |
|---|---|---|
| `config/car/laser_mount.yaml` | car bringup, simulator | `base_link`→`laser`: today f1tenth's defaults 0.27, 0, 0.11 m, [MEASURE] on the car |
| `config/car/camera_mount.yaml` **[TBD]** | car bringup, simulator | `base_link`→`camera_link` |
| `config/lidar/sl450.yaml` | simulator | SL450 model: 40 Hz, 270°, 0.2°, ranges 0.05–25 m, noise, dropout |
| `config/maps/*.yaml` | simulator | track maps for the simulator |
| `maps/` | SLAM, AMCL | saved room maps |
| `polimi_bringup/config/mux.yaml` **[Proposed]** | mux | priorities, timeouts (0.2 s), locks |
| `polimi_bringup/config/` slam, amcl, ekf **[Proposed]** | logic | node parameters |
| `vesc.yaml` (`f1tenth_stack`) | VESC driver | gains, servo offset, wheelbase |
| `joy_teleop.yaml` (`f1tenth_stack`) | joystick teleop | button map: LB enables `/teleop`; RB's `autonomous_control` gates nothing today (OI-09) |

# 4 Interfaces

## 4.1 The car interface

This table is the contract the stack is built against; it becomes the README of `polimi_bringup` **[Proposed]**. All message types are standard ROS ones, so no interface package is needed. Exact camera and IMU topic names follow `orbbec_camera`'s defaults, so the car needs no remaps **[TBD: T-02]**.

Only `/ackermann_cmd` goes from the stack to the hardware; every other row comes from the hardware.

| Topic / frame | Type | Car | Simulator |
|---|---|---|---|
| `/ackermann_cmd` | `AckermannDriveStamped` | `ackermann_to_vesc` | actuation model |
| `/teleop` | `AckermannDriveStamped` | `joy_teleop` | keyboard bridge |
| `/joy` | `Joy` | `joy_node` | stand-in **[TBD: T-01]** |
| `/scan`, frame `laser` | `LaserScan` | `orbbec_camera` LiDAR | LiDAR model |
| `/odom` | `Odometry` | `vesc_to_odom` | odometry model |
| TF `odom`→`base_link` | TF | `vesc_to_odom` | odometry model |
| camera image, info | `Image`, `CameraInfo` | `orbbec_camera` | camera model (`TDD_Gym`) |
| IMU | `Imu` | `orbbec_camera` | IMU model (`TDD_Gym`, deferred until OI-08) |

| Topic / frame | Type | Car | Simulator |
|---|---|---|---|
| TF `base_link`→`laser`, `camera_link` | static TF | `car.launch.py` | `polimi_sim` |

`/drive` (`AckermannDriveStamped`) is inside the stack, not part of the car interface: any controller writes it, and it enters the mux at priority 10.

### 4.2 Path following

- **Input path:** `nav_msgs/Path` on one topic, any frame; offline paths latched, online paths recomputed per camera frame (FR-PTH-002, 005).
- **Input pose:** TF lookup from the path's frame to `base_link` (FR-PTH-003).
- **Output:** `/drive`. A stale online path stops the car (FR-PTH-007).
- **Controllers:** the PID and Pure Pursuit, interchangeable (FR-PTH-004). Paths are closed loops only (FR-PTH-005).
- **Tuning:** each controller publishes its lateral error, heading error and look-ahead point as topics for plotting; every gain is a live parameter (FR-PTH-009).

## 5 Security & Privacy

- **Physical safety comes first:** the mux, its locks and AEB (FR-SAF) are the safety design; they run on the car and in the simulator unchanged (DEC-03).
- **Network:** on Mac and PC the container's ports bind to 127.0.0.1 only. The car's container uses host networking; ROS traffic on the car's network is not authenticated **[TBD]**.
- **Secrets and personal data:** none in the stack. Camera images may show people in the room; nothing is recorded by default.
- **Repository access:** protected branches and required reviews, as in `CONTRIBUTING.md`.

## 6 Requirements Traceability

| Requirement | Component(s) | Test(s) |
|---|---|---|
| `FR-DRV-001` | `vesc_driver`, `vesc_ackermann` | bench: `ros2 topic echo /odom` while driving by joystick |
| `FR-DRV-002` | `joy`, `joy_teleop` | bench: `/teleop` only while the enable button is held |
| `FR-DRV-003` | `orbbec_camera` | bench: rates of `/scan`, image, IMU |
| `FR-DRV-004` | `car.launch.py` (DEC-09) | `tf2_echo base_link laser` matches `laser_mount.yaml` |
| `FR-DRV-005` | calibration runbook in `docs/software/` | the procedure run on the car; its result in `camera_mount.yaml` |
| `FR-MAP-001`, `002` | `slam_toolbox` | AC-V1-01 |
| `FR-LOC-001`, `002` | AMCL (default, OI-07) | AC-V1-01 |
| `FR-LOC-003` | EKF (DEC-10) | **[TBD: OI-08]** |
| `FR-LIN-001`, `002` | `polimi_lines` | recorded camera data; AC-V1-02 |
| `FR-PTH-001` to `009` | `polimi_control` | unit tests on the error and window search; error topics while driving; AC-V1-02, 03 |

| Requirement | Component(s) | Test(s) |
|---|---|---|
| `FR-SAF-001` to `005` | `ackermann_mux`, `mux.yaml`, `polimi_safety` | AC-V1-04, in the simulator first |
| `FR-SIM-001` to `003` | car interface, `hw:=sim` | `ros2 topic info` on both sides; AC-V1-02 |
| `FR-DOC-001` | runbooks, RASD, TDD | review by someone outside the team |
| `FR-ADV-001` to `003` | **[V2]** **[TBD: OI-01]** | AC-V2-A |
| `FR-ADV-011` to `013` | **[V2]** **[TBD: OI-01]** | AC-V2-B |
| `NFR-POR-001` | images, DEC-01, DEC-11 | the same launch on `arm`, `x86`, `jet` |
| `NFR-VAR-001`, `002` | logic | laps on more than one layout and room |
| `NFR-VAR-003` | `polimi_lines` | **[TBD: OI-06]**, once floor and tape are known |
| `NFR-PER-001` | logic, controller | laps at 3 m/s (AC-V1-02) |
| `NFR-PER-002` | `orbbec_camera` LiDAR | bench: `ros2 topic hz /scan` |
| `NFR-PER-003` | `orbbec_camera` camera | bench: `ros2 topic hz` on the image, $\geq$ 30 fps; 60 fps once 1280×720 at 60 in YUYV is verified |
| `NFR-PER-004` | `orbbec_camera`, `vesc_driver` | **[TBD]**: method defined with the car's measurement session |
| `NFR-PER-005` | `mux.yaml`, VESC | stop sending `/drive`: the mux drops it after 0.2 s and the VESC timeout stops the motor (0.5 s in the simulator, the VESC app value on the car) [VERIFY] |
| `NFR-SAF-001` | mux, locks, AEB | AC-V1-04, in the simulator and on the car |
| `NFR-SAF-002` | test plan | AC-V1-04 passed on the car before autonomous laps |
| `NFR-MNT-001`, `002` | development standards (DEC-12) | PR review |
| `FR-OFF-001` | **[TBD: OI-11]** | optional |

# 7 Implementation

## 7.1 Technology Stack

| Layer | Choice | Why |
|---|---|---|
| Compute | Jetson Orin Nano Super, 500 GB NVMe | BOM; fits the F1TENTH deck |
| OS, container | Ubuntu 24.04 images, Docker; JetPack 7.2.1 on the car, GPU through the NVIDIA Container Toolkit **[TBD: T-10]** | DEC-01, DEC-11, DEC-13 |
| Middleware | ROS 2 Jazzy, `rmw_zenoh_cpp` | DEC-01 |

| Layer | Choice | Why |
|---|---|---|
| Drivers | `f1tenth_system` (`vesc`, `ackermann_mux`, `teleop_tools`), `orbbec_camera` | C-022 |
| Mapping, localization | `slam_toolbox`, `nav2_amcl`, `robot_localization` | scope; DEC-02 |
| Visualization | Foxglove bridge; RViz through noVNC | runs without a display on the host |
| Python | system Python of the image; `uv` on the host | house rule |

### 7.2 Plan

1. **Jazzy migration** (section 7.3): Jazzy images, the JAX gym, the simulator fork as a submodule.
2. **`polimi_bringup`:** the car interface README, `stack.launch.py`, `sim.launch.py`, `common.launch.py`, our `mux.yaml`; `simcar` calls `hw:=sim`; with DEC-08 the mux leaves the simulator fork.
3. **Safety:** OI-09 locks and AEB, with the `/joy` stand-in (T-01), tested in the simulator.
4. **Logic:** mapping and localization, line detection (on replayed camera bags of the real tape, `TDD_Gym`), path following (PID and Pure Pursuit, error topics), as the base platform needs.
5. **Car bringup**, when the car is built: `car.launch.py`, udev rules, measured configs, the camera calibration procedure, safety tests on the car.
6. **Advanced topic** [V2], once OI-01 is decided.

### 7.3 Jazzy migration

Moves every image to ROS 2 Jazzy, the simulator to the JAX gym, and `polimi_sim` into the simulator fork. Four steps, each checked before the next; the distro, the gym and the frames change in separate commits, so a failure points to one change.

1. **Image:** every image on `ros:jazzy`, `f1tenth_system` on `jazzy-polimi`, the DEC-02 packages, and the same gym and bridge as today, so a breakage is about ROS only. Checks: the images build for any host UID (Ubuntu 24.04 images ship a user with UID 1000); `cv2` and NumPy load from the gym's pip copies (DEC-02); the packages and drivers build; today's simulator runs.
2. **Simulator in the fork:** `polimi_sim` moves into `f1tenth_gym_polimi` with its history and is ported to the JAX gym and the `dev-jazzy` bridge, with labelled default configs and a config overlay (`TDD_Gym`). The mux stays in the fork until DEC-08.
3. **Submodule:** the fork becomes `ros2_ws/src/f1tenth_gym_ros`, `polimi_sim` leaves this repo, and the measured car configs stay in `ros2_ws/config/`. Steps 2 and 3 merge together. Checks: the workspace layout; on the new gym, the scan matches the gym's beam by beam and physics keeps $\geq$ 95 % of 100 Hz on the Mac; the config overlay logs each file's source.
4. **GitHub:** default branches and rules move to the Jazzy branches; the Humble images and volumes are removed after acceptance.

Rollback: the Humble images, volumes and branches stay until step 4, so going back is a revert and the old image tags.

### 7.4 Development Standards

- **Branches and reviews:** every change through a PR. Protected branches (`main`; the forks' `humble-*`, `jazzy-*`, `dev-humble*`, `dev-jazzy*`) block force pushes and deletion and need one approval.
- **Submodules:** changing a fork takes a commit there, then a pointer commit here; `push.recurseSubmodules=check` blocks a pointer to an unpushed commit. `scripts/setup_git.sh` sets this up and adds the `upstream` remotes.
- **Upstream sync:** merge upstream into our branch, never rewrite it; preview with `git log ours..upstream`.
- **Pins:** gym and bridge refs are bumped on purpose, never by following a branch (NFR-MNT-002).
- **Rebuilds:** code and configs need only `cb`; a `package.xml`, the Dockerfile or a pin needs an image rebuild.
- **Python:** `uv` on the host, never `pip install` directly.

## 8 Testing

### 8.1 Strategy

| Level | Scope | Tooling / gate |
|---|---|---|
| Unit | Controller math, line fitting, AEB time to collision | `colcon test`, pytest |
| Contract | The car interface on both sides | `ros2 topic info` and `tf2_echo`: same names, types, frames on `hw:=car` and `hw:=sim` |
| System, simulator | Full flows: map, localize, laps, safety | AC-V1-01 to 04 in the simulator first |
| System, car | The same flows on the car | AC-V1-01 to 04 on the car; safety before autonomous laps (NFR-SAF-002) |

### 8.2 Acceptance Criteria

As in `RASD_F1tenth`: AC-V1-01 to 04 for the base platform, AC-V2-A or AC-V2-B for the advanced topic. Leaving the lines is judged by the people watching on the car, and measured in the simulator.

## 9 Open Items

| ID | Topic | Notes |
|---|---|---|
| T-01 | Deadman and e-stop in the simulator | Docker Desktop on macOS has no USB passthrough, so there is no joystick in the container. The locks need a stand-in for `/joy`: a keyboard or Foxglove key that holds the deadman and toggles the emergency stop, built with the OI-09 locks (Plan, step 3). |
| T-02 | Camera and IMU topic names [VERIFY] | From `orbbec_camera`'s Gemini launch, once the camera is on the bench; the simulator's camera model uses the same names. |

| ID | Topic | Notes |
|---|---|---|
| T-03 | LiDAR frame name [VERIFY] | From `orbbec_camera`'s `lidar.launch.py`; set it to `laser` in `car.launch.py` if it differs. |
| T-04 | Confirm DEC-03 to DEC-10 | DEC-08 is decided to follow the Jazzy migration (8 Nov 2026). The others are still Proposed; with DEC-06, `RASD_F1tenth` C-022 is reworded. |
| T-05 | Wheelbase | `vesc.yaml` has 0.25 m, the simulated odometry the gym's 0.3302 m. Measure the car and align both. |
| T-06 | Mount measurements [MEASURE] | `laser_mount.yaml` holds f1tenth's defaults; the camera mount comes from the calibration procedure (FR-DRV-005). |
| T-07 | VESC device | udev rule on the Jetson for `/dev/sensors/vesc`, expected by `vesc.yaml`. |
| T-08 | `orbbec_camera` version | 2.9.3 is in the Jazzy apt repo; check it with the SL450's firmware at car bringup. |
| T-09 | Unmatched remap | Today's bringup remaps `ackermann_cmd_out`→`ackermann_drive`, which matches nothing: the mux publishes `ackermann_cmd`. Gone with DEC-06. |
| T-10 | CUDA in the container [VERIFY] | With JetPack 7.2.1 (DEC-13), check at bringup how CUDA reaches the `ros:jazzy` container through the NVIDIA Container Toolkit: mounted from the host, or installed in the `jet` image. |
| T-11 | Battery connector [BOM GAP] | The VESC has XT90; the BOM has no TRX (Traxxas High-Current) to XT90 adapter. Check the Zeee 3S packs' connector (section A.3). |
| T-12 | Steering limits [MEASURE] | `vesc.yaml`'s servo clamp gives +0.31 rad left and −0.26 rad right (minimum turning radius about 1.0 / 1.2 m); the simulator clamps at ±0.4189 rad (`TDD_Gym`). Measure on the car; the simulator then takes the car's limits. |

# A Hardware Selection

The trade studies behind the picks in `RASD_F1tenth` C-010, C-012 and C-013. Prices are the ones seen on the date of each study; the master BOM holds the current vendors and prices.

## A.1 LiDAR (study of 23 Sep 2026)

> **Pick: Orbbec Pulsar SL450** (C-012). For F1TENTH the scan rate matters far more than range: at 8 m/s a 10 Hz LiDAR lets the car move 80 cm between scans, which makes scan matching and localization much harder. The SL450 is the only candidate that matches the Hokuyo UST-10LX, the standard F1TENTH LiDAR (40 Hz, 0.25°, 270°, about EUR 1,200).

The study was written for a track bounded by walls. The track is now tape on the floor (RASD C-001 to C-005): the LiDAR cannot see it and maps the room, so "track" about range or SLAM reads "room". The pick is unchanged.

| Model | Price | Rate | Res. | Range 70–90% / 10% | FOV | Weight | Interface, principle |
|---|---|---|---|---|---|---|---|
| Orbbec Pulsar SL450 | EUR 531 | **40 Hz** | 0.2° | 45 / 15 m | 270° | 320 g | Ethernet; dToF, 72 kHz, IP65 |
| Richbeam Lakibeam 1L | EUR 549 | 30 Hz | 0.25° | 40 / 20 m | 270° | 160 g | Ethernet, USB-C; dToF, 45 kHz, 9–36 V |
| Richbeam Lakibeam 1 | EUR 439 | 30 Hz | 0.25° | 25 / 15 m | 270° | 160 g | Ethernet, USB-C; dToF, 43.2 kHz, 9–36 V |
| Richbeam Lakibeam 1S | EUR 340 | 20 Hz | 0.5° | 15 / 10 m | 270° | 160 g | Ethernet, USB-C; dToF, 6–36 V |
| Slamtec RPLIDAR S3 | EUR 549 | 20 Hz | 0.225° | 40 / 15 m | 360° | 115 g | UART 1 Mbps; dToF, 5 V |
| LSLIDAR N301 | EUR 1,100 | 20 Hz | 0.36° | 30 m / – | 360° | 406 g | Ethernet; ToF, IP67 |
| Slamtec RPLIDAR A2M12 | EUR 227 | 15 Hz | ∼0.225° | 12 m / – | 360° | 190 g | UART 256 kbps; triangulation |

All seven from RobotShop EU, with a ROS 2 driver; accuracy ±2–3 cm for the ToF units, 1–2.5% of range for the A2M12.

| Scan rate | At 5 m/s | At 8 m/s |
|---|---|---|
| 10 Hz | 50 cm | 80 cm |
| 20 Hz | 25 cm | 40 cm |
| 30 Hz | 17 cm | 27 cm |
| 40 Hz | 12.5 cm | 20 cm |

- **SL450:** the only one at 40 Hz (rates 15, 20, 25, 30, 40 Hz), and 0.2° even at 40 Hz thanks to its 72 kHz sample rate; multi-echo noise filtering. Against: 320 g, twice the Lakibeams, and new (March 2025), so less community troubleshooting than Slamtec.
- **Lakibeam 1, the fallback:** 30 Hz at 0.25° for EUR 439, 160 g, runs straight from a 3S LiPo; Richbeam's ROS 2 driver publishes `/scan`. The 1L only adds range we do not need.
- **Rejected for speed:** Lakibeam 1S (0.5° at 20 Hz), RPLIDAR S3 (capped at 20 Hz), LSLIDAR N301 (twice the price, the heaviest, 20 Hz), RPLIDAR A2M12 (15 Hz, 12 m, slow UART).
- **Field of view:** 360° gives nothing on an F1TENTH car: the chassis and electronics block the rear, so 270° is the norm.

## A.2 Camera (study of 7 Oct 2026)

> **Pick: Orbbec Gemini 335L** (C-013), EUR 420 from OpenElab, DigiKey Italy at EUR 408 as a second source. The Intel RealSense D435i stays in the BOM as an unselected alternative.

The LiDAR cannot see the tape, so a camera finds the lines (RASD FR-LIN). What that asks of the camera, most important first:

| Need | Why |
|---|---|
| Global-shutter colour | At 5–8 m/s a rolling shutter skews and smears the lines, worst in corners. The tape colour is not decided, so colour matters. |
| Wide field of view, $\geq$ 60 fps | Both lines stay in view near the car and through corners. At 8 m/s, 30 fps is 27 cm per frame. |
| Accurate pixel-to-floor projection | Good factory intrinsics, and the camera's pitch and roll to the floor at every frame: braking and accelerating pitch the car, so a fixed mount calibration is not enough at speed. |
| Depth aligned to colour, and an IMU | To measure that pitch and roll each frame: fit the floor plane in the depth image, or use the IMU. |
| Hardware timestamps | 10 ms of timestamp error at 8 m/s is 8 cm (NFR-PER-004). |
| USB 3, light | The Orin Nano's only Ethernet port goes to the LiDAR; a 1/10 car. |

| # | Camera | Price | Notes |
|---|---|---|---|
| 1 | Orbbec Gemini 335L / 336L | EUR 420 (335L) | Global-shutter colour 1280×800 up to 60 fps, 94°×68°; depth aligned to colour in the camera; IMU up to 1 kHz; depth from 0.17 m, 95 mm baseline; 133 g, IP65. The 336L adds an IR-pass filter on the depth cameras. |
| 2 | RealSense D436 | EUR 576 | The same capabilities (1280×800 at 60, 90°×65°, aligned depth, IMU) in the D435i's body, same driver and topics. |
| 3 | Arducam B0385 (OV9782) plus an IMU board | EUR 69 + ∼EUR 25 | Global-shutter colour only: no depth, no factory calibration, no hardware IMU sync. USB 2, so 100 fps only as MJPEG. |
| 4 | Luxonis OAK-D W | ≈EUR 522 | Global-shutter colour, 127°, can run a segmentation network on the camera. Strong fisheye distortion, weaker depth, rare in F1TENTH. |
| 5 | RealSense D435i | EUR 405 | Rolling-shutter colour, 1080p at 30 fps, 69°×42°. Usable only if the tape shows on its global-shutter IR cameras. |

- **Skipped:** Gemini 335/336 (short baseline), OAK-D Lite, OAK-D S2 (rolling-shutter colour); ZED Mini and ZED 2i (rolling shutter, depth on the Orin's GPU; the global-shutter ZED X needs a GMSL2 card); RealSense D455 (bigger, from EUR 536, minimum depth 0.4–0.5 m misses the floor in front of the car).
- **Colour-camera offset:** the colour camera sits beside the left IR camera, the depth origin: 23.75 mm along $x$ on the 335L; the IMU is at (7.866, 1.068, −14.248) mm. Both are factory transforms and the floor calibration absorbs them. Mount the colour lens on the car's centreline.
- **Projection:** fit a floor plane to the depth image rather than read depth at each tape pixel; stereo depth on a floor seen at a shallow angle is noisy.
- **335L over 336L:** the 336L's IR-pass filter darkens the IR images under LED room light with the projector off, so the 335L is safer if those images are ever used alone.

## A.3 Chassis (study of 24 Sep 2026)

> **Pick: Traxxas Ford Fiesta ST Rally VXL, 74276-4** (C-010), bought complete (Plan A). It looks end of life [VERIFY]: Traxxas shows no price and UK retailers list it as discontinued. Plan B, if no complete car can be found, is a rolling chassis from Traxxas spare parts.

The F1TENTH (RoboRacer) build guide is written for the Traxxas Slash 4x4 Premium (brushed Titan motor, XL-5 ESC). Applied to the Rally VXL's parts list (REV. 260209):

| Guide step (Slash 4x4) | On the Rally VXL |
|---|---|
| Remove the body clips and body | Clipless body (7420, 7414): body and trim discarded. |
| Remove the XL-5 ESC | Remove the VXL-3s ESC (3355R); the VESC 6 MkVI replaces it. |
| Remove the receiver box, receiver, antenna tube | 7424, TQi receiver 6533, 1726 and 1926 discarded: the joystick and the Jetson replace the radio. |
| Replace the brushed motor | Not needed: the Velineon 3500 brushless (3351R) is kept; the BOM's 4 mm-to-3.5 mm bullet adapters connect it to the VESC. |
| Keep the servo | The 2075 servo is kept; the VESC drives it through the PPM cable. |
| Remove the battery hold-down | 7426 discarded; the battery sits under the platform deck. |
| Remove the nerf bars, reuse their M3 screws | No nerf bars on the Rally [VERIFY]; take M3 screws from the BOM's kit. |
| Three 45 mm M3 standoffs into the chassis | Chassis 7422 ("Slash LCG and Rally"): check the hole pattern on the real part [VERIFY]. |
| Battery to VESC through TRX to XT90 | Not in the BOM **[TBD: T-11]**. |

**Kept** (the rolling chassis, motor and servo): chassis tub, bulkheads, tie bars, skidplate, gear cover; the 4WD driveline (sealed front and rear diffs, centre slipper clutch, centre driveshaft, half-shafts, spur and pinion, motor mount and plate); the suspension (arms, pins, caster and steering blocks, stub-axle carriers, shock towers, camber and toe links, Ultra Shocks); the steering (bellcranks, servo saver, drag link, toe links, 2075 servo); wheels, tyres, hubs, bearings; the Velineon 3500.

**Discarded:** body and accessories (7427-x, 7420, 7414, 7410, 7413X, 7419); receiver box, seals and plug (7424, 7425, 3698); battery hold-down (7426); antenna tube (1726, 1926); VXL-3s ESC, BEC and fan (3355R, 2260, 3340); TQi radio and telemetry (6509R, 6528, 6533); the box's batteries and tools.

### A.3.1 Plan B parts

Universal Traxxas part numbers, sold in EU shops under the same number (prefix TRX or TXX). [VERIFY] marks a variant or colour not confirmed as stock; check them, and every quantity, against the exploded views of the 74276-4.

| Qty | Part | Description | Notes |
|---|---|---|---|
| *Chassis and structure* | | | |
| 1 | 7422 | Chassis, Slash LCG / Rally | platform-deck standoffs go here [VERIFY] |

| Qty | Part | Description | Notes |
|---|---|---|---|
| 1 | 7430X | Bulkhead, front | |
| 1 | 7429 | Bulkhead, rear | |
| 1 | 6823 | Bulkhead tie bars, front and rear, aluminium | colour [VERIFY] |
| 1 | 7435 | Front skidplate, angled | |
| 1 | 7477 | Gear cover with wire retainer | guards the spur gear against cables |
| 1 | 7434 | Foam body bumper, low profile | recommended |
| *Driveline* | | | |
| 1 | 6788 | Front differential, complete (13/37, rally) | |
| 1 | 6789 | Rear differential, complete (13/37, rally) | |
| 1 | 6878 | Slipper clutch, complete | a centre slipper, not a centre diff |
| 1 | 6893R | Slipper input shaft, bearing adapter, pin | may come with 6878 [VERIFY] |
| 1 | 6888 | Centre rear drive hub | |
| 1 | 6888X | Front drive hub, hardened steel | |
| 1 | 7455 | Centre driveshaft, plastic | [VERIFY]; 6855 aluminium is the upgrade |
| 2 | 7450 | Front half-shaft, Extreme HD, 6 mm axle | sold as left or right |
| 2 | 7451 | Rear half-shaft, Extreme HD, 6 mm axle | sold as left or right |
| 1 | 6842 | Spur gear, 0.8 metric pitch | stock tooth count **[TBD]** (50, 52 or 54T) |
| 1 | – | Pinion, 32-pitch steel, 3 mm bore | stock tooth count **[TBD]**; the smaller stock pinion suits indoor speeds |
| 1 | 7460A | Motor mount, 3x6 flat-head screw, nylock nut | |
| 1 | 7490 | Motor plate | colour [VERIFY] |
| *Suspension* | | | |
| 2 | 6731 | Suspension arms, front/rear L&R (2 per pack) | 4 arms in total |
| 1 | 9042 | Suspension pin set, Extreme HD | 9042X is the hardened upgrade |
| 1 | 9032 | Caster blocks, Extreme HD | colour [VERIFY] |
| 1 | 9037 | Steering blocks, Extreme HD, L&R | colour [VERIFY] |
| 1 | 9050 | Rear stub-axle carriers, Extreme HD | colour [VERIFY] |
| 1 | 9038 | Front shock tower, Extreme HD | |
| 1 | 9039 | Rear shock tower, Extreme HD | |
| 1 | 7432 | Camber links, 49 mm (72 mm c-c) | front/rear [VERIFY] |
| 1 | 7431 | Camber links, 49 mm (63 mm c-c) | front/rear [VERIFY] |
| 1 | 7433 | Toe links, 47 mm (77 mm c-c), rear | [VERIFY] |
| 1 | 3760A | Ultra Shocks, long, with springs (front pair) | [VERIFY]; rebuild kit 2362 |
| 1 | 3762A | Ultra Shocks, XX-long, with springs (rear pair) | [VERIFY] |
| 1 | 5529 | Shim set, hollow balls | |
| 1 | 2742 | Rod ends, long, hollow balls | |
| *Steering* | | | |
| 1 | 2075 | Digital high-torque waterproof servo | kept by the build |
| 1 | 6845X | Bellcranks, servo saver, spring, retainer, horn | |
| 1 | 7438 | Drag link | 6845A aluminium is the upgrade |
| 1 | 7439 | Front toe links, composite, hollow balls | |

| Qty | Part | Description | Notes |
|---|---|---|---|
| 1 | 2545 | Bellcrank bushings, 5x8x2.5 mm | |
| *Wheels and hubs* | | | |
| 2 | 7473 | Tyres and wheels, glued, gravel (2 per pack) | gravel grips poorly indoors; road tyres **[TBD]** |
| 1 | 9069 | Wheel hubs, 12 mm hex, Extreme HD | |
| 1 | 2754 | Stub-axle pins | |
| 1 | 3647 | Wheel nuts, 4 mm flanged nylock | [VERIFY]; 1747 is the alternative |
| 1 | 5854 | Hub retainer, 17 mm, M4 | used on this model [VERIFY] |
| *Motor* | | | |
| 1 | 3351R | Velineon 3500 brushless, 3.5 mm bullets | kept; the 3350R combo is easier to find, then drop its ESC |

- **Bearings** (blue rubber-sealed, 2 per pack), quantities **[TBD]** from the exploded views: 5116 (5x11x4 mm), 5117 (6x12x4), 5118 (8x16x5), 5119 (10x15x4), 5120 (12x18x4), 1985 PTFE washers (5x8x0.5).
- **Screws and nuts** (hex drive, packs of 6), quantities **[TBD]**: button-head 3x6 to 3x40 (2575–2580, 2582, 2583, 2591, 2592) and 4x10/12/14 (3936–3938); cap-head 2.5x6, 2.5x8, 3x8, 3x15 (3215, 3965, 1552, 2586); flat-head 3x6, 3x8, 3x15 (3932, 3931, 3646); countersunk 3x6, 3x12, 4x12, 4x15 (2534, 2552, 2542, 2546); nylock nuts 3 and 4 mm (2745, 1747); grub screws (2743); washers 3x6 (2746); shock shoulder screws (3642X). One missing screw stops the build: this is Plan B's biggest risk.
- **Tools:** spare parts come without the box's hex keys; add a hex driver set (Traxxas 2748R or 8712, or 1.5/2.0/2.5 mm hex and 4/7 mm nut drivers).
- **Cost:** Plan B heads toward or above the price of the complete car before the motor, servo, diffs, shocks, bearings and screws are priced, and it adds a full assembly. Plan A stays the choice.